

Program: Diploma in Renewable Energy	
Course Code: 5299A	Course Title: Control Systems Lab
Semester: 5	Credits: 1.5
Course Category: Program Elective course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To familiarize with control system applications
- To analyze the first and second order system using time domain analysis
- To lay the foundation for development of MATLAB/SIMULINK model for various control systems.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge of Electrical circuits		Fundamentals of Electrical and Electronics Engineering	1
Force and Torque		Applied Physics II 2	2
Basic knowledge in Mathematics		Engineering Mathematics I & II	1 & 2

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Implement control systems in various applications	9	Applying
CO2	Analyze the first and second order systems using time domain analysis	9	Applying

CO3	Model a control system by its transfer function using Matlab/Simulink or any similar software	9	Applying
CO4	Determine the impulse, step, and ramp response of a system, given a transfer function model and analyze its time domain specifications	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Implement control systems in various applications		
M1.01	Analyze the performance of ON/OFF controller to control the temperature of an oven.	3	Applying
M1.02	Simulate the A.C position control system.	3	Applying
M1.03	Capture the Speed – Torque characteristics of DC servomotor	3	Applying
CO2	Analyze the first and second order systems using time domain analysis		

M2.01	Explore the time domain analysis of first order systems when subjected to a unit step input.	3	Applying
M2.02	Perform the time domain analysis of the second order system when subjected to a unit step input.	3	Applying
M2.03	Simulate the performance characteristics of a DC motor speed control system.	3	Applying
	Lab Test – I	3	
CO3	Model a control system by its transfer function using Matlab/Simulink or any similar software		
M3.01	Model the transfer function of an LTI systems using Matlab or any similar software	3	Applying
M3.02	Prepare the transfer function block in SIMULINK or any similar software	3	Applying
M3.03	Simulate the response of mass/spring/damper system using SIMULINK	3	Applying
C04	Determine the impulse, step, and ramp response of a system, given a transfer function model and analyze its time domain specifications		
M4.01	Plot the step, ramp, impulse response of first order system using MATLAB	3	Applying
M4.02	Plot the step, ramp, impulse response of second order system using MATLAB	3	Applying
M4.03	Analyze the damping in a second order system using MATLAB.	3	Applying
M4.04	Investigate the time Domain specifications of Under damped second order system using MATLAB.	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	A NagoorKani, Control Systems, RBA Publications
T2	Nagarath I. J. and Gopal M., Control System Engineering, 5/e, New Age Publishers
T3	Ogata K, Modern Control Engineering, 5/e, Prentice Hall of India.
R1	ImthiasAhamed T. P, Control Systems, Phasor Books, 2016
R2	Gopal M., Control Systems Principles and Design, 4/e, Tata McGraw Hill.

Online Resources

Sl.No	Website Link
1	https://plctt-coep.vlabs.ac.in/exp/transmitter-level-control/simulation.html Website Link
2	https://nptel.ac.in/courses/107106081
3	https://www.youtube.com/watch?v=bZNKURutM3g
4	https://da-iitb.vlabs.ac.in/exp/water-level-control/simulation.html
5	https://www.youtube.com/watch?v=uK15lhFMXdE&list=PLBBGwliUvNPDYNygN-GkVgHcEaQsrH8nC
6	http://homepages.ed.ac.uk/jwp/control06/controlcourse/course/map/level/index.html

Student Activity**Suggested Open-ended Experiments:**

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on an open-ended experiment of their choice.